



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

Machine Learning for Prediction of Catalytic Properties

AI FOR CHEMISTRY — CH457

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Abstract

Traditional heterogeneous catalyst discovery relies heavily on empirical trial-and-error, which is bottlenecked by the immense combinatorial complexity of material compositions and operating conditions. To transition toward data-driven catalyst selection, this study evaluates five machine learning architectures - Random Forest, XGBoost, CatBoost, TabPFN, and TabICLv2 - for predicting the reaction features of methanol dehydrogenation and formic acid decomposition. Utilizing the CatTestHub database (derived from the CPEC), the models were tasked with predicting absolute kinetic reaction rates and product selectivities. Systematic k -fold cross-validation demonstrated that the state-of-the-art transformer-based tabular foundation models, TabPFN and TabICLv2, significantly outperform traditional tree-based ensembles, achieving a stable RMSE below 3.0 to 4.0 $\mu\text{mol}_{\text{product}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{min}^{-1}$ in small-data regimes. Model evaluation was performed by benchmarking several tabular regression models for reaction rate and selectivity prediction under cross-validation. The analysis compared single-target, target-wise multi-output, and native joint-output strategies using sMAPE to handle outputs with different numerical scales. Multi-target predictions revealed that simpler models perform better in native joint-output than more complex models, which can be explained by the small size of the dataset. Furthermore, training the architectures on a merged dataset demonstrated that foundation models are highly resilient to cross-reaction complexity, showing minimal performance degradation. Finally, a sequential Bayesian Optimization campaign was executed around the high-performance design space of formic acid decomposition, utilizing TabICLv2 as a probabilistic surrogate. The algorithm successfully identified an optimized catalytic operating window, demonstrating the viability of foundation models to guide experimental exploration and accelerate rational catalyst design, while highlighting the need for broader datasets.

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Introduction

Heterogeneous catalysis underpins over 80% of global manufacturing, though material discovery has historically relied on costly, empirical trial-and-error [1]. Machine learning (ML) addresses this by transforming experimental data into models that navigate chemical complexity and identify non-linear correlations inaccessible to human intuition [2]. This study utilizes the CatTestHub database, which is derived from the CPEC, incorporating environmental parameters (T, P) alongside catalyst composition to ensure model generalizability [3]. We implement a rigorous benchmarking framework to compare architectures ranging from gradient boosting to state-of-the-art TabICLv2 [4].

The central objective of this study is to implement and evaluate machine learning architectures for the prediction of catalytic performance in model reactions, specifically methanol dehydrogenation and formic acid decomposition. Methanol dehydrogenation serves as a vital pathway for sustainable hydrogen generation, and formic acid decomposition is a cornerstone of liquid organic hydrogen carrier (LOHC) technology. The activation energy (E_a) of these reaction often fluctuates between 50 kJ mol^{-1} to 110 kJ mol^{-1} , a range highly sensitive to the specific metal-support interface and surface morphology [5, 6]. This results makes computational models very costly and difficult to implement. While traditional Density Functional Theory (DFT) has long been the standard for modeling these energy barriers, its precision has historically been limited by functional-specific errors. However, current state-of-the-art hybrid functionals now achieve a Mean Absolute Error (MAE) of approximately 6.3 kJ mol^{-1} [7]. Furthermore, as of 2026, the integration of descriptor-based foundation models and graph neural networks has pushed machine learning precision toward the "chemical accuracy" threshold of 1 kcal mol^{-1} (4.2 kJ mol^{-1}) [8]. By benchmarking these advanced models against experimental data, this research seeks to bridge the gap between high-fidelity computational predictions and the practical requirements of industrial catalysis.

To reach this benchmark, the project systematically compares hold-out versus cross-validation techniques, encoding strategies for catalyst descriptors, and single versus multi-target prediction frameworks. Finally, we implement Bayesian Optimization strategies.

1 Supervised ML algorithms for the prediction of one reaction feature - holdout method

Figure 1 presents the initial performance of the five selected models (TabPFN, TabICLv2, CatBoost, RandomForest, and XGBoost) for predicting the methanol dehydrogenation reaction rate ($\mu\text{mol}_{\text{product}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{min}^{-1}$). This initial evaluation was conducted after pretreatment of the data, using a single partition method (holdout method).

A clear hierarchy is observed among the algorithms: TabPFN and TabICLv2 significantly dominates the benchmark with the lowest error (RMSE=4.05-4.1), followed by CatBoost (RMSE= 6.55). Conversely, the RandomForest and XGBoost models exhibit markedly higher errors, exceeding 14 and 19 RMSE points, respectively. Similar trends can be seen with sMAPE.

From a chemical perspective, the superiority of TabPFN and TabICLv2, which serve as "foundation models" for tabular data, is due to its ability to generalize effectively on modest-sized datasets, which are typical of experimental catalysis. The solid performance of CatBoost compared to other ensemble methods (XGBoost/RF) suggests that its native handling of categorical variables is a major asset for encoding catalyst descriptors (metal types, supports). The high error of XGBoost here might indicate an increased sensitivity to experimental data noise or a need for finer hyperparameter tuning than what is provided out-of-the-box.

Considering that in this dataset, reaction rate values range between 0.14 and $255 \mu\text{mol}_{\text{product}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{min}^{-1}$, direct analysis of the performances of this model do vary importantly with the reference value. Indeed the prediction on the activation energy depends on the relative ratio of expected rate and measured rate.

2 Implementing k- cross-validation method to enhance performances

The implementation of k-fold cross-validation allows for a more refined estimation of model robustness. Figure 2 illustrates the evolution of the mean RMSE for the reaction rate as a function of the number of splits. An

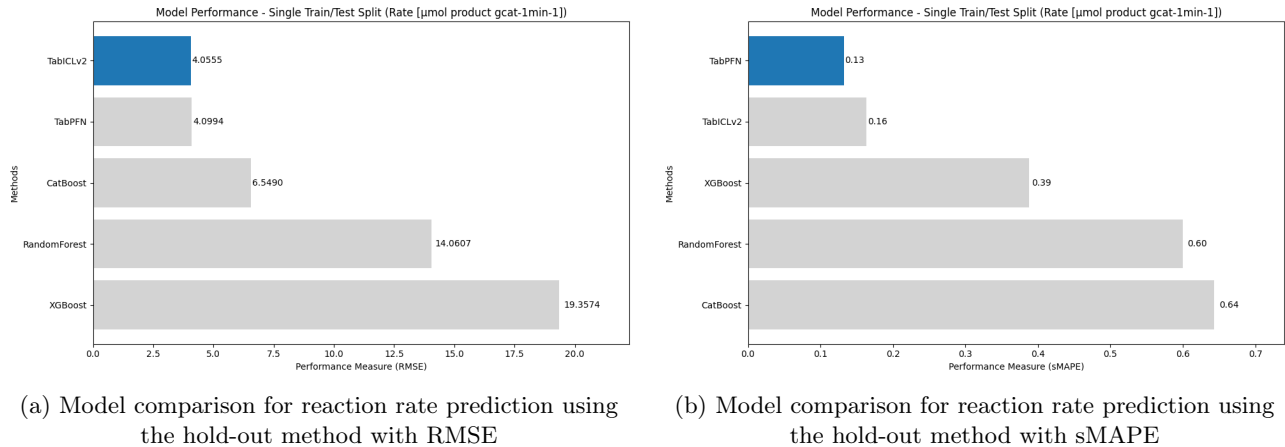


Figure 1: Performance of the different models on methanol dehydrogenation reaction rate prediction using the holdout method, for both metrics

overall downward trend in error is observed as k increases: for all models, performance improves and stabilizes beyond 10-15 splits. TabPFN, closely followed by TabICLv2, maintain its leading position, with a mean error stable 2 to 4 times lower than that of other models. Indeed, TabPFN, respectively TabICLv2, reaches $\text{RMSE} < 3.0$, respectively $< 4.0 \mu\text{mol}_{\text{product}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{min}^{-1}$ for $k > 10$, while the gap between CatBoost and RandomForest progressively narrows. It also to be noticed that no change of order was observed between the models and that increasing k from 3 to 10 allowed for a 3-4 $\mu\text{mol}_{\text{product}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{min}^{-1}$ RMSE decrease in each model. In other words the gap between different models, though not constant, did not evolve significantly.

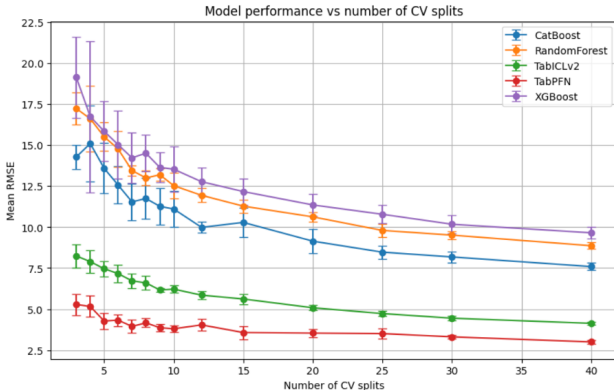


Figure 2: Influence of the number of splits on the performance of the different models for methanol dehydrogenation reaction rate prediction ($\mu\text{mol}_{\text{product}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{min}^{-1}$)

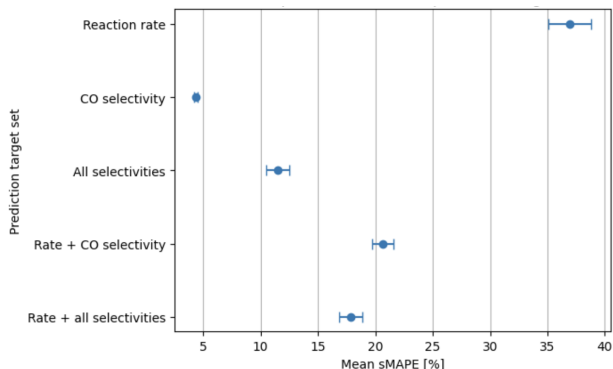
Analysis of CO selectivity (Fig. 7, see Appendix) confirms that mean RMSE decreases for all models as k increases. TabICLv2, closely followed by TabPFN, consistently lead and reach an RMSE below 3% at $k = 20$ ($k = 25$ for TabPFN), while ensemble models converge higher (3.5%–4.2%). Crucially, tree-based methods (Random Forest, XGBoost, and CatBoost) exhibit significant volatility at low k values (standard deviation up to 0.75), whereas TabPFN and TabICLv2 remain stable with a standard deviation below 0.3.

Although larger training sets are needed to capture complex catalytic behaviors, high k values increase overfitting risks on this small dataset ($n \approx 100$). Tree-based models tend to capture experimental noise, as evidenced by their irregular variability (Fig. 7b). In contrast, TabPFN’s transformer architecture, pre-trained for Bayesian inference, resists the memorization of specific data points. Consequently, $k = 10$ provides the optimal balance between learning non-linear correlations and ensuring valid statistics. TabPFN’s superior stability confirms its advantage for high-dimensional catalysis over traditional ensemble methods.

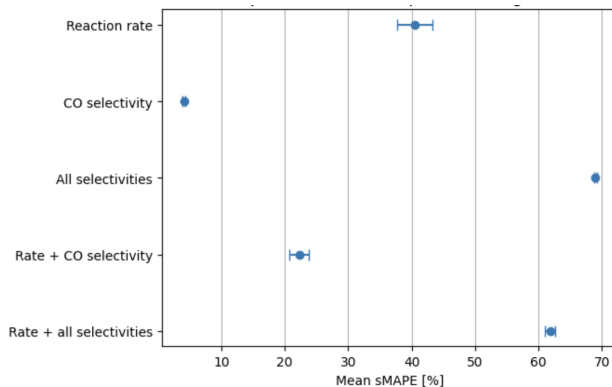
3 Predicting multiple features at a time

To evaluate whether the performance of a catalyst can be predicted beyond a single output, five target configurations were tested: reaction rate, CO selectivity, all selectivities, rate + CO selectivity, and rate + all selectivities (Figure 3). These outputs have different units and numerical ranges; we consequently used sMAPE at 20-fold cross validation to evaluate performance.

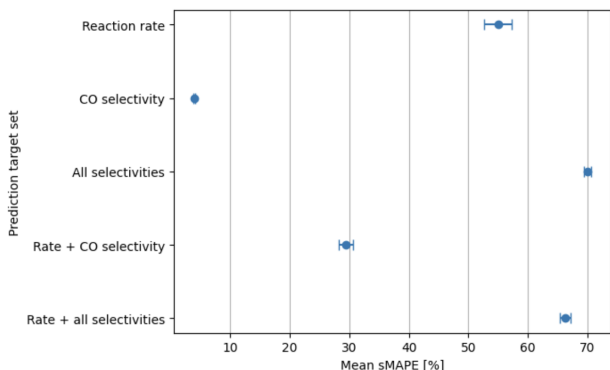
Two multi-output strategies were considered. First, a target-wise wrapper was used, where one independent regressor is trained for each output and the final score is obtained by averaging the errors across targets. This approach allows comparison between different target combinations, but it does not learn correlations between reaction rate and selectivities. TabPFN was not included in this multi-output benchmark because, in our implementation, it is designed for single-target regression and does not directly support multiple outputs. Therefore, the wrapper benchmark was restricted to RandomForest, XGBoost, CatBoost, and TabICLv2.



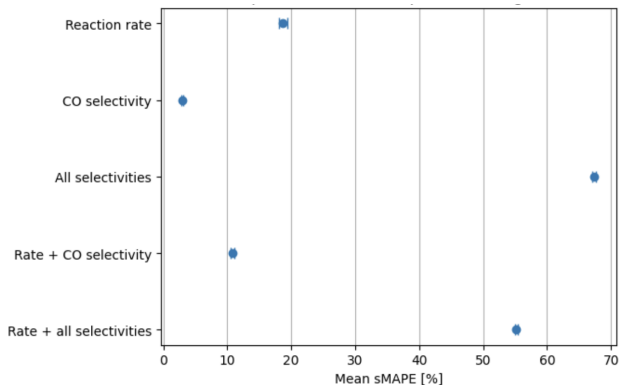
(a) Performances of RandomForest across prediction targets.



(b) Performances of XGBoost across prediction targets.



(c) Performances of CatBoost across prediction targets.



(d) Performances of TabICLv2 across prediction targets.

Figure 3: Performances of the different models across varied prediction outputs combinations (target-wise approach).

The model comparison (Figure 3) show that single selectivity target output is easier to predict than reaction rate. This is consistent with the nature of the data: selectivities are bounded percentage values, whereas reaction rates span a wider numerical range and are more sensitive to experimental conditions. However, this trend does not extend to the full selectivity vector. When all selectivities are predicted together, the error increases substantially for most models and is often higher than for reaction rate alone. This suggests that the difficulty is not selectivity as a single scalar target, but the simultaneous prediction of several heterogeneous and competing product selectivities.

Reaction rate is generally easier to predict than the full set of selectivities for most models, likely because all-selectivity target set combines outputs with different ranges, frequencies, and noise levels. Minor or low-valued selectivities are especially difficult under sMAPE, because the denominator depends on the magnitude of the actual and predicted values; therefore, small absolute deviations can translate into large percentage [9].

$$\text{sMAPE} = \frac{100}{n} \sum_{i=1}^n \frac{2|y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i|}$$

The comparison between native multi-output models and target-wise wrapper models (Figure 10) shows that selectivity-only targets are generally easier to predict than target sets including the reaction rate. However, the target-wise wrapper does not learn correlations between outputs, since each target is predicted independently. Native joint-output regression, used as a complementary benchmark to test true multi-output prediction, show that RandomForest performs best, likely because its ensemble averaging behavior provides robustness on a small and heterogeneous dataset.

4 Algorithms trained on two different reactions

As to determine whether the models can be used on different reactions simultaneously without affecting that much the overall performances on predicting the reaction rate, the models were trained on a merged dataset of methanol dehydrogenation and formic acid decomposition catalytic performances. In these two datasets, similar catalysts were used under similar operating conditions.

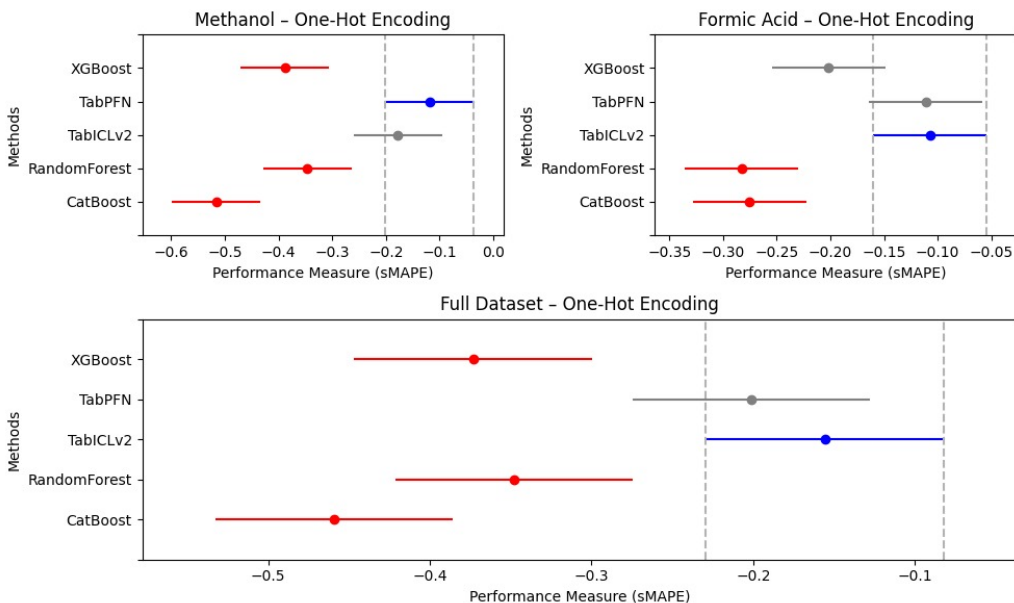


Figure 4: Performance of the different models on the reaction rate depending on the training dataset (20 folds)

Figure 4 displays the models prediction results for the rate of reaction for single reactions and both reactions training. Comparing the single-reaction benchmarks with the "Full Dataset" results reveals that the foundation models (TabPFN and TabICLv2) exhibit high resilience to cross-reaction training. While TabICLv2 achieves its peak performance on Formic Acid (sMAPE ≈ 0.11), its error on the merged dataset (sMAPE ≈ 0.16) remains significantly lower than the individual performances of traditional algorithms like XGBoost or Random Forest (sMAPE ranging from 0.3 – 0.5 across training sets).

It is also noticeable that if TabPFN outperforms TabICLv2, trained on methanol dehydrogenation, both models have similar performances on the formic acid decomposition dataset. However it is only on the merged dataset that TabICLv2 becomes clearly the best model, matching the expected performances of the TabArena benchmark (see Fig. 6).

This stability of these two models indicates that the models successfully navigate the combined chemical space without suffering from significant interference between the two reaction datasets. In contrast, while tree-based models such as CatBoost maintain stable error levels across the merge, they fail to reach the precision threshold required for reliable industrial generalizability.

5 Implementation of Bayesian Optimization (BO) techniques

Seeing the performance of TabICLv2 on the reaction rate, we can now run a Bayesian optimization (BO) campaign aiming to maximize the reaction rate of one of the two reactions (Formic Acid Decomposition), using TabICLv2 to predict the rate of the proposed point for each iteration, as running the experiments ourselves is unrealistic for the scope of this project.

We restrict the search to a design space constructed around the ten best experiments in the dataset. Indeed, in a realistic experimental setting, chemists would focus on the region where the system shows good performance. The ten best points therefore serve to delimit what is deemed to be the promising part of the space. For the categorical variables, all ten best experiments use the same metal (Palladium), the support (Carbon), and inert gas (Helium), so the categorical part of the design space is fixed. Likewise, the H_2 fraction is zero for all ten best experiments, so it is fixed. For the other continuous variables (temperature, pressure, WHSV, and the inlet mole fractions of formic acid and CO_2), we take the minimum and maximum values observed among the top ten and expand each end of the interval by an arbitrary $\Delta = (\max - \min)/3$, making sure the bounds are still physically meaningful. Five evenly-spaced points are selected for each of those intervals. The resulting design space has the same feature order and encoding as the original dataset on which TabICL has been trained, so it can process it directly.

BO is initialized with five points selected using a max-min distance strategy to ensure good coverage of the restricted space. The same seed is used for both BO and random search, so they start with the same initial point. BO relies on a Gaussian Process surrogate to choose the next point, while random search simply samples uniformly across the unexplored space. In both cases, TabICL is used to simulate the outcome of each proposed experiment, which allows us to run the optimization loop. The result of this run is given on Figure 5.

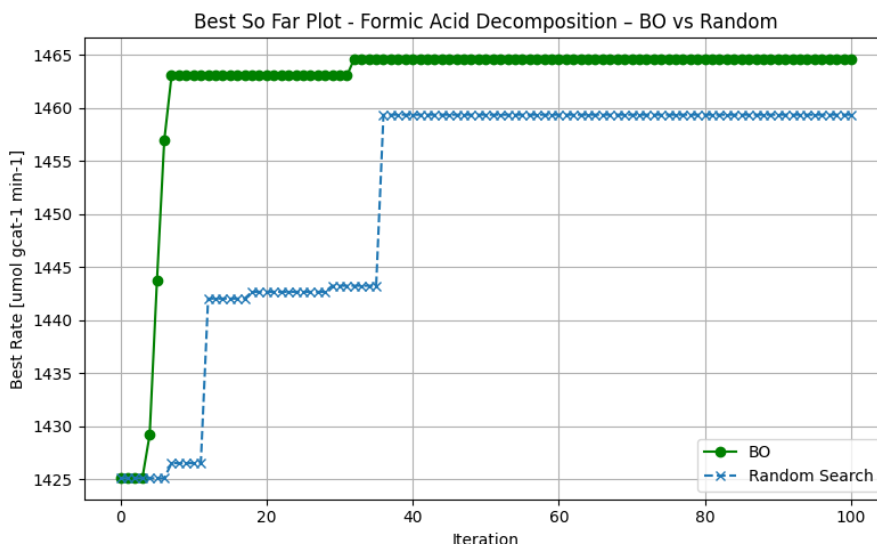


Figure 5: Best-So-Far Plot for Formic Acid Decomposition rate, comparing BO and random search.

BO is more efficient than random search, finding points within the defined design space with higher reaction rates in less iteration. Moreover, the BO converges on a design point with a predicted reaction rate of $1465 \mu\text{mol g}_{\text{cat}}^{-1} \text{min}^{-1}$, which is higher than the highest rate predicted by TabICLv2 on the original experimental points of $1457 \mu\text{mol g}_{\text{cat}}^{-1} \text{min}^{-1}$. This means the model has identified conditions that it considers more promising than what is present in the dataset already.

This however represent only a minor increase (0.5% increase), and exploring other design spaces could yield better results. Indeed, if the true optimum is outside the explored discrete region, it cannot be found. The categorical space is also fixed to a single combination, so alternative catalyst systems cannot be discovered. In addition, the optimization reflects the behavior of TabICLv2, so the identified optimum is model-dependent. For example, the highest predicted rate by TabICLv2 is actually lower than the measured rate with the same experimental parameters.

Conclusion

The objective of this project was to benchmark different models against experimental data in heterogeneous catalysis. Five machine learning architectures - Random Forest, XGBoost, CatBoost, TabPFN, and TabICLv2 - were evaluated to predict the reaction rates and product selectivities of methanol dehydrogenation and formic acid decomposition. Throughout the evaluations led, the state-of-the-art tabular foundation models, TabPFN and TabICLv2 significantly outperformed traditional tree-based ensembles, showcasing a superior capacity to generalize within small-data regimes ($n \approx 100$) typical of experimental heterogeneous catalysis. Furthermore, a sequential Bayesian Optimization (BO) campaign was successfully implemented around the high-performance design space of formic acid decomposition to maximize the reaction rate. It successfully identified an optimized catalytic operating window. However, the localized nature of this optimum highlights clear model dependency and the geometric constraints of a restricted design space.

While this study confirms that transformer-based foundation architectures can effectively map multi-reaction spaces, accelerating catalyst discovery requires significantly broader, high-dimensional experimental datasets to allow for robust cross-reaction extrapolation.

6 Appendix

6.1 Methods

6.1.1 Models Used

In this project, five machine learning models were evaluated: Random Forest, CatBoost, XGBoost, TabPFN and TabIClv2. These models cover a range of approaches, from classical ensemble methods to modern foundation models for tabular data.

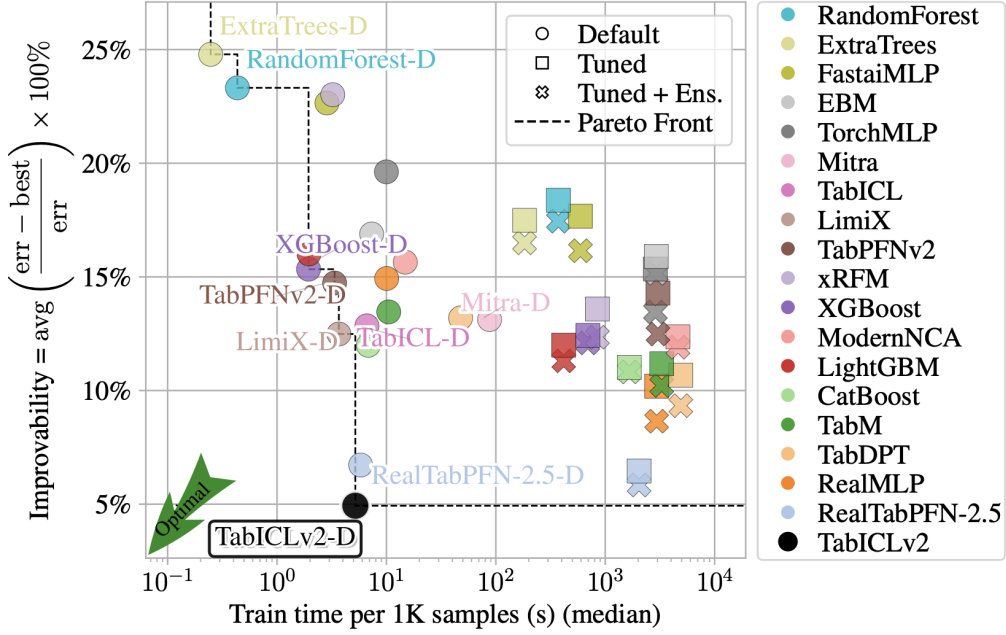


Figure 6: Performances of the different models based on the TabArena benchmark (improvability vs train time)[10]

Random Forest: Random Forest is an ensemble learning method based on decision trees, introduced by Breiman [11]. It constructs multiple decision trees using bootstrapped samples of the data and random subsets of features, and combines their outputs through majority voting (classification) or averaging (regression). The method is widely used for both classification and regression tasks due to its robustness and ability to model nonlinear relationships. Theoretical analysis has shown that Random Forests can achieve consistency under certain assumptions and can adapt well to high-dimensional settings [12].

XGBoost: XGBoost (Extreme Gradient Boosting) is an optimized implementation of gradient-boosted decision trees introduced by Chen and Guestrin. Unlike bagging methods such as Random Forest, XGBoost builds trees sequentially, where each new tree corrects the errors of the previous ones [13]. The model incorporates regularization, efficient handling of sparse data, and system-level optimizations such as parallel computation [13]. These design choices make XGBoost highly scalable and one of the most widely used algorithms for structured data problems, as demonstrated in benchmarking studies [14].

CatBoost: CatBoost is a gradient boosting algorithm designed to handle categorical features efficiently. It introduces techniques such as ordered boosting and specialized encoding strategies to reduce prediction shift and prevent target leakage [15]. A key advantage of CatBoost is its ability to process categorical variables directly without extensive preprocessing. This simplifies the modeling pipeline and often leads to strong out-of-the-box performance. Benchmark studies show that CatBoost achieves competitive or state-of-the-art performance compared to other gradient boosting methods on a wide range of tabular datasets [14].

TabPFN: TabPFN (Tabular Prior-Data Fitted Network) is a transformer-based model specifically designed for tabular data. Developed by Hollmann in 2025, it is pre-trained on a large collection of synthetic datasets and performs inference on new datasets without additional training. The model approximates Bayesian inference in a single forward pass using a transformer architecture, eliminating the need for hyperparameter tuning and significantly reducing computation time. TabPFN achieves highly accurate predictions on small tabular datasets and often matches or outperforms classical machine learning methods without requiring task-specific training [16].

TabICLv2: TabICLv2 is an open-source tabular foundation model that serves as a faster and more scalable successor to the TabPFN architecture [17]. Introduced by Qu et al. in 2026, it leverages the principles of In-Context Learning (ICL) to perform highly efficient inference on new datasets without requiring task-specific training or hyperparameter tuning. By optimizing the underlying transformer-based mechanisms, TabICLv2 overcomes the computational and memory constraints often associated with previous foundation models.

6.1.2 Dataset preparation

The raw catalytic dataset was first cleaned by systematically removing empty rows and columns. Given the high complexity inherent to heterogeneous catalyst surfaces and the limited number of readily available descriptors for heterogeneous catalysts, this study relies exclusively on this database.

A subset of seventeen primary features was selected, spanning catalyst formulation parameters—such as active metal type, support identity, weight loading, active component molecular weight, active site density, and dispersion—and operational conditions, including weight hourly space velocity (WHSV), reaction temperature, total pressure, and specific inlet feed compositions (mole fractions of inert gas, CO, CH₄, and methanol). Categorical descriptors like metal and support types were processed using ordinal transformers rather than nominal encodings to preserve feature density without artificially expanding the dimensionality of this specialized dataset.

6.1.3 Choice of the metric:

In order to assess the accuracy of our regression models, we have chosen to use the root mean squared error (RMSE) whenever possible and when relevant the sMAPE metric. Whereas RMSE measures the absolute magnitude of error, sMAPE is a percentage-based metric (0–200%) that provides relative error.

The choice of RMSE as a basis is motivated by its ability to penalise large errors more heavily due to the squared term, which is essential in the context of catalysis where a significant discrepancy in the prediction of a reaction rate or selectivity can distort the interpretation of a material’s performance.

However, given that our study involves target variables with very different units and orders of magnitude—such as reaction rates and selectivity percentages—it becomes also very relevant to compare relative errors. Especially, the prediction on the activation energy depends on the relative ratio of expected rate and measured rate. Therefore for reaction rate and multiple features comparison, the sMAPE metric was chosen.

6.1.4 Performance criterion on reaction rate predictions

Determination of the activation energy: In this section our protocol to determine the energy of activation of the model reactions (methanol dehydrogenation and formic acid decomposition) is described. Due to their stoichiometry, both of the reactions can be accounted to be of order 1. Therefore, using the Arrhenius law and working at constant reactant partial pressure, their reaction rate can be written as :

$$r = k \exp\left(-\frac{E_a}{RT}\right) \quad (1)$$

To determine the kinetic parameters E_a (activation energy) and A (pre-exponential factor), the equation is

linearized by taking the natural logarithm:

$$\ln(r) = \ln(k) - \frac{E_a}{R} \left(\frac{1}{T} \right) \quad (2)$$

Then, using a linear regression, the activation energy can be determined. For instance, in the methanol dehydrogenation set, activation energies were found within the range of 40-80 kJ/mol.

Activation energy sensitivity to reaction rate and criterion definition: This section aims at determining if the predictions of our models based on reaction rates are relevant when compared to literature data. The objective is to be able to determine if the performances of the model are acceptable or not.

For a measured rate r_{meas} and a predicted rate r_{pred} , corresponding energies can be described using the following relation :

$$\Delta E = E_{a_{meas}} - E_{a_{pred}} = RT \ln\left(\frac{r_{pred}}{r_{meas}}\right) \quad (3)$$

According to the literature, acceptable for ΔE are below 4.2 kJ mol^{-1} [8]. This set is using operating temperatures between 450 and 500 K for methanol dehydrogenation and of 373 K for formic acid decomposition. At these temperatures, using the precedent equation, this threshold is reached for values of the $\frac{r_{pred}}{r_{meas}}$ ratio of 3. Such ratio is equivalent to a sMAPE value of 1.00.

So for the prediction of our models to be considered acceptable, we must have $\text{sMAPE} < 1$.

6.1.5 Multi-output prediction strategy

Two multi-output strategies were implemented. The first strategy used a target-wise wrapper based on independent regressors. For multi-target configurations, one model was trained separately for each output, and the reported score corresponds to the average sMAPE across targets. This approach is useful for benchmarking target-set difficulty, but it should not be interpreted as true correlation learning between outputs.

The second strategy used native joint-output regression, where one model is trained directly on the full target matrix. This approach was tested for RandomForest, XGBoost, and CatBoost. CatBoost was configured with a MultiRMSE loss. TabPFN and TabICLv2 were not included in the native joint-output benchmark because their implementations in this study did not provide direct support for multi-target regression.

6.2 Complementary figures

6.2.1 Implementing k- cross-validation method to enhance performances: CO selectivity

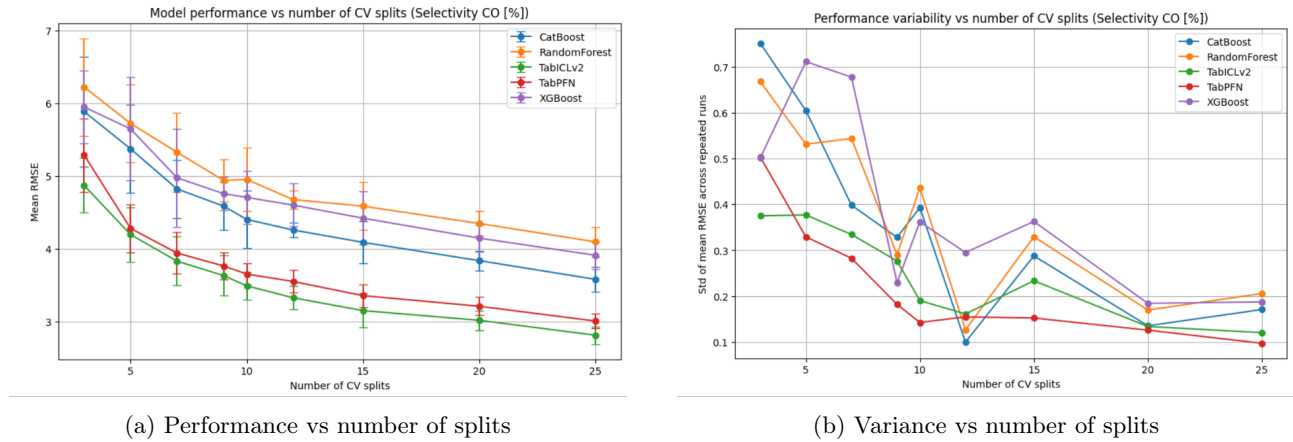


Figure 7: Influence of the number of splits on the performance and variance of the different models for CO selectivity prediction.

6.2.2 Implementing k- cross-validation method to enhance performances: CO₂ and CH₄ selectivity

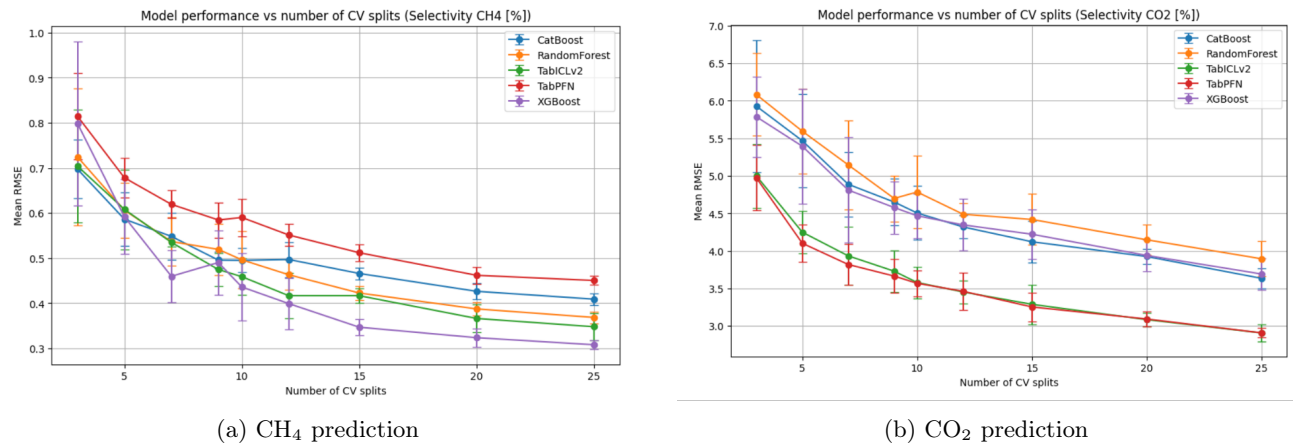


Figure 8: Influence of the number of splits on the performance of the different models for CH₄ and CO₂ selectivity prediction.

6.2.3 Models performances

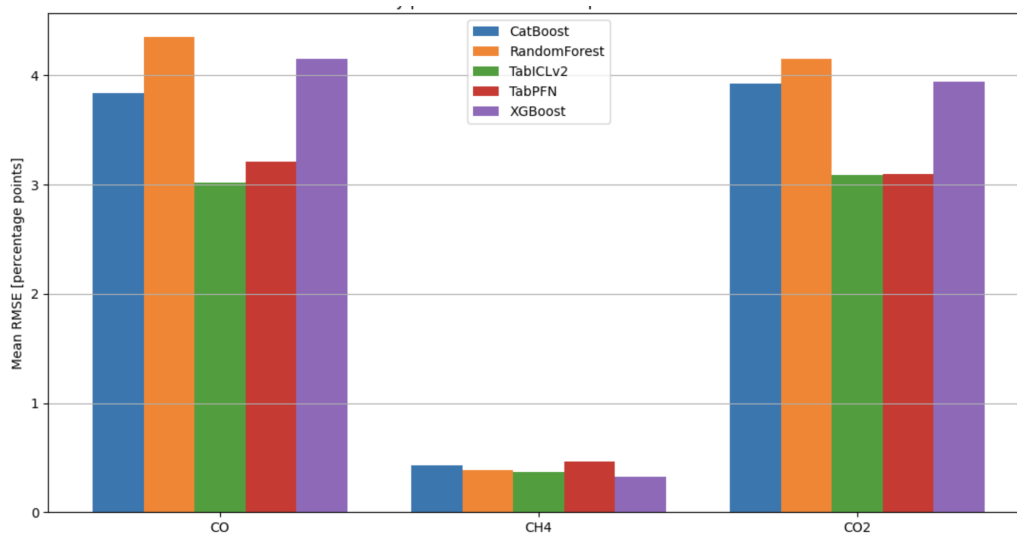


Figure 9: Performances of the different models in the prediction of different single-output selectivity targets.)[10]

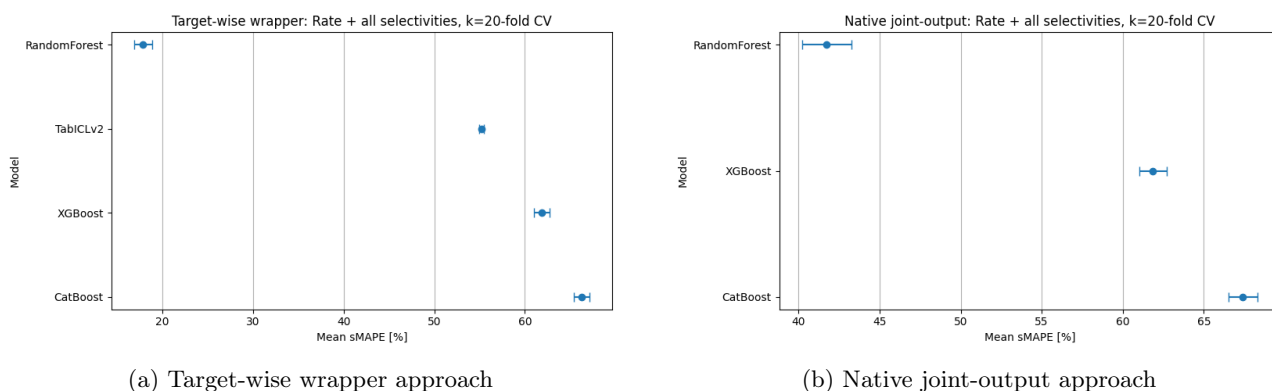


Figure 10: Comparison of target-wise wrapper and native joint-output strategies for predicting reaction rate and selectivities.

To test joint-output learning, a benchmark was performed using native multi-output models. In this setting, a single model predicts several outputs simultaneously. TabICLv2 was excluded from this benchmark because native joint-output regression was not supported in our implementation. Among the tested models, RandomForest gave the best native joint-output performance 10b. This is likely due to its robustness on small datasets: by averaging many decorrelated trees, it reduces variance and is less sensitive to noisy or heterogeneous targets than more complex boosting models. Overall, multi-output prediction is relevant for catalytic screening, but its benefit is model- and target-dependent, while also being heavily influenced by the size of the training data set.

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